University of Idaho
Department of Computer Science

CS 4622/5622
Applied Data Science with Python

Dr. Alex Vakanski

Lecture 1

A Short History and Current State of Artificial Intelligence

(not required for quizzes or assignments)

Lecture Overview

- Artificial Intelligence vs. Machine Learning vs. Deep Learning vs. Data Science
- AI approaches and goals
- Evolutionary context
- Timeline of Artificial Intelligence
- Current AI capabilities
  - DL success in Computer Vision
  - DL success in Natural Language Processing
  - Large Language Models
- Modern AI systems
  - Foundation models
  - Reasoning models
  - AI Agents
- AI limitations and challenges
- Prospective trends in AI

Artificial Intelligence

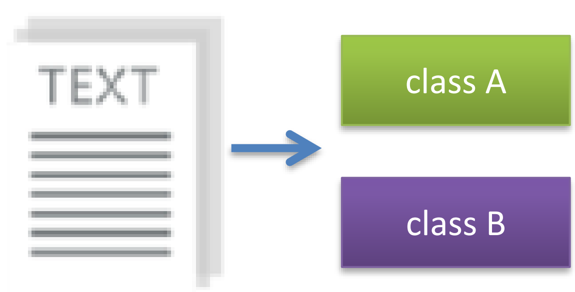
AI vs. Machine Learning vs. Deep Learning vs. Data Science

- ***Artificial Intelligence (AI)*** is a scientific field concerned with the development of algorithms that allow computers to reason or learn without being explicitly programmed
- AI as an academic discipline was founded in 1956
- AI studies theories and technologies related to:
  - Planning and reasoning
  - Knowledge representation
  - Machine learning
  - Natural language processing
  - Perception
  - Motion and manipulation

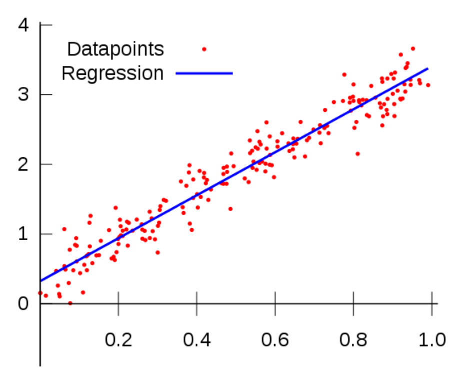
Machine Learning

AI vs. Machine Learning vs. Deep Learning vs. Data Science

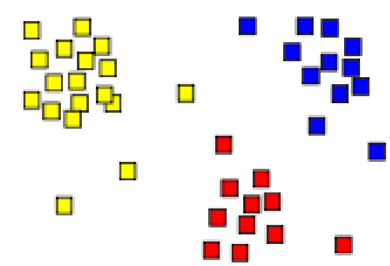
- ***Machine Learning (ML)*** is a subfield of Artificial Intelligence, that studies methods that learn from data and make predictions on unseen data
- Categories of ML approaches
  - Supervised learning: learning with labeled data
    - Example: image classification, email classification
    - Example: regression for predicting real-valued outputs
  - Unsupervised learning: discover patterns in unlabeled data
    - Example: cluster similar data points
  - Reinforcement learning: learn to act based on feedback/reward
    - Example: learn to play Go or Minecraft



Classification



Regression

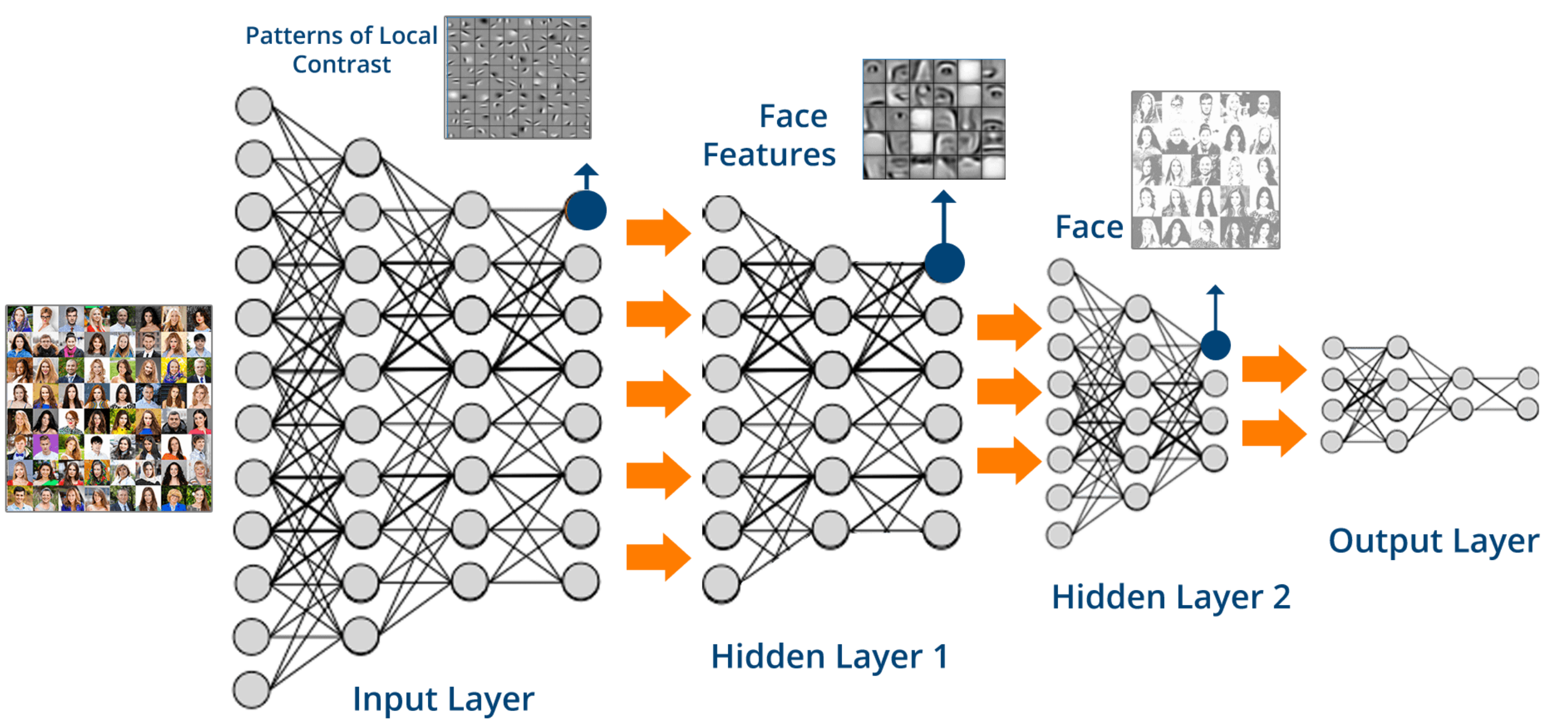


Clustering

Deep Learning

AI vs. Machine Learning vs. Deep Learning vs. Data Science

- *Deep Learning (DL)* is a sub-area in Machine Learning that uses artificial neural networks (ANNs) with multiple layers for learning data representations
  - Advantages of DL: automatically extract features in data, process complex high-dimensional data, are scalable with data, model size, and computational power
  - Some of the most common architectures in deep ANNs include multi-layer perceptron NNs, convolutional NNs, recurrent NNs, graph NNs, transformer NNs

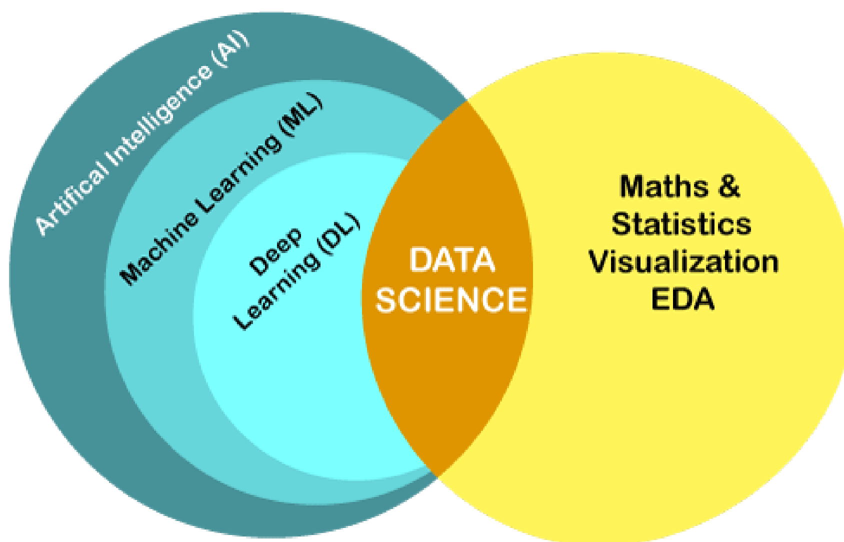


Data Science

AI vs. Machine Learning vs. Deep Learning vs. Data Science

- ***Data Science (DS)*** is an interdisciplinary field that uses scientific methods and algorithms to extract knowledge from data, and applies the insights to application domains (such as to make business decisions)
- Data Science versus Machine Learning
  - DS has overlaps with ML (as well as with AI and DL): DS can rely on ML approaches, but it can also obtain insights from data via mathematical and statistical analysis, data visualization techniques, exploratory data analysis (EDA), or other approaches that don't necessarily require training an ML model

AI vs. ML vs. DL vs. DS



What is Intelligence?

AI Approaches and Goals

- An intelligent agent is a system that perceives the environment and takes actions to maximize the chances of achieving its goals
  - Goals can vary, e.g., human goals can be to make a coffee, build a wall, solve a math problem, drive a car, cook a meal, etc.
- A widely used **definition**: ***Intelligence*** is an agent's ability to achieve goals in a <u>wide range of environments</u>
- Intelligent agents are able to acquire and retain knowledge, and use it to respond effectively to new tasks or act in new situations and environments
  - E.g., more intelligent humans should be able to solve physics problems that they haven't seen before (e.g., think of Einstein)
- Intelligence encompasses many related abilities for:
  - Reasoning and rational thinking, comprehending ideas, applying planning, problem-solving
  - Learning and adaptation, dealing with unexpected situations and uncertainties
  - Interacting with the real world to perceive, understand, and act

How to Develop Intelligent Machines?

AI Approaches and Goals

- AI scientists in 1950s believed that machines with human-level intelligence can be developed within 10 to 20 years
- *Initial AI approaches* included:
  - Imitate step-by-step reasoning that humans use to solve a problem
  - Create a knowledge database based on human domain knowledge about a task, and develop an inference engine to process the states and make decisions
- Challenges: handling uncertainties, combinatorial explosion (the space of solutions quickly becomes too large for most problems)
  - These approaches failed to deliver, as the scientists underestimated the complexity of human intelligence
- Various misconceptions about intelligence have perpetuated in the AI field
  - E.g., computers can process information -> human thinking is similar to logic processing -> encoding human thinking into a program can lead to intelligent machines
  - E.g., chess is a game of intellect and chess players are very intelligent people -> developing computers that can reason and play chess at a human expert level can lead to machines with human-level intelligence

Weak vs. Strong AI

AI Approaches and Goals

- AI systems are commonly categorized based on their capability into weak and strong AI systems
- ***Weak AI***, or ***narrow AI***: designed and trained to solve one specific task
  - E.g., image classification ML models
  - E.g., Deep Blue computer that defeated the world chess champion
- ***Strong AI***, or ***artificial general intelligence (AGI)***: capable of learning and performing a variety of intellectual tasks that a human can
  - AGI includes competence across many unfamiliar domains
  - At present, AGI systems do not exist: current systems exceed human performance on some narrow tasks, but fail at other tasks that most people find trivial
    - Predictions for achieving AGI vary widely among experts, ranging from within 5 years to more than 50 years
    - there isn't an agreed test for when one would
- ***Artificial super intelligence (ASI)***: surpasses human intelligence across all domains
  - ASI is a hypothetical future AI that vastly outperforms humans on all tasks

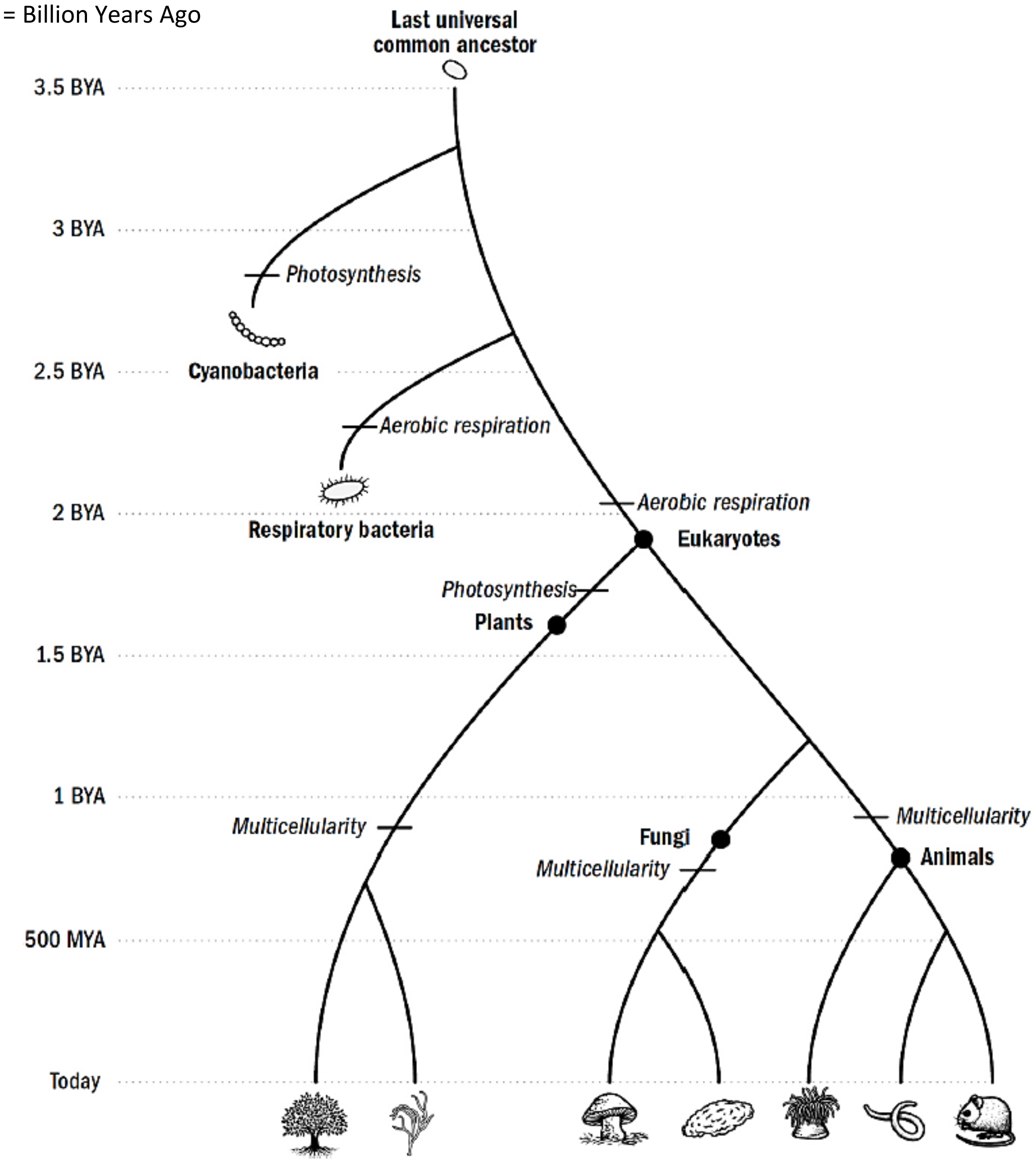
Timeline of Biological Intelligence

Evolutionary Context

- Earth formed about 4.5 billion years ago (BYA)
- The last universal common ancestor to all life on Earth appeared about 3.5 BYA
- The first simple plants developed about 1.5 BYA
- Multicellular organisms, such as algae and fungi, evolved about 1 BYA

Tree of Life

BYA = Billion Years Ago



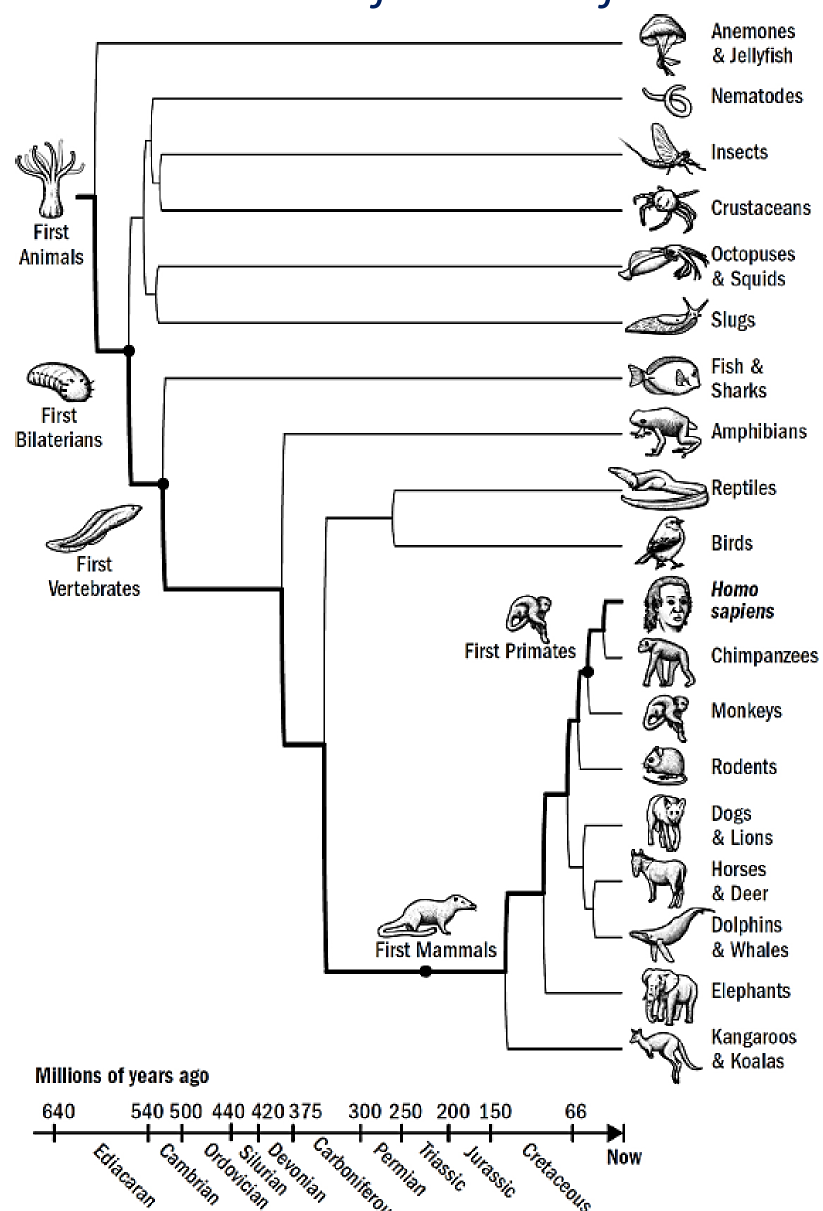
Source: Max Bennett (2023) A Brief History of Intelligence

Timeline of Biological Intelligence

Evolutionary Context

- The first animals evolved around 650 million year ago (MYA)
- The Cambrian explosion (540 MYA) is a period of large animal diversity, with species developing hard shells and skeletons
- First mammals appeared around 250 MYA, and primates evolved around 66 MYA
- Animals exhibit intelligent behavior: they learn and adapt, and plan their actions to achieve goals

Evolution of Animal Life



Source: Max Bennett (2023) A Brief History of Intelligence

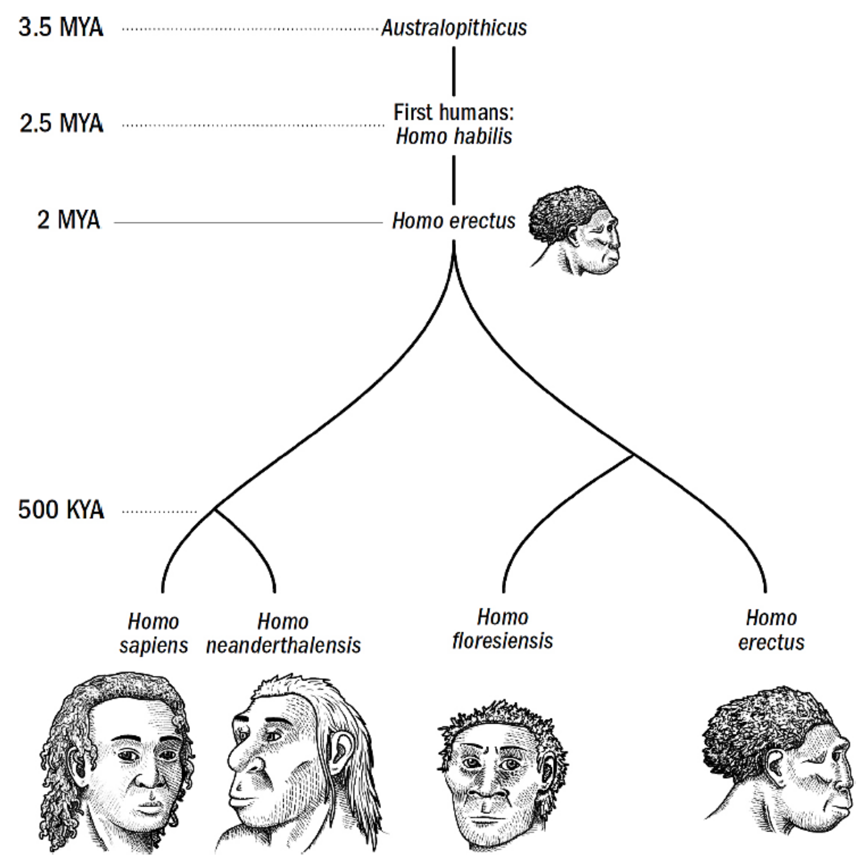
Timeline of Biological Intelligence

Evolutionary Context

- Early humans inherited brain structures and features of intelligence from our evolutionary ancestors
  - The human brain is a scaled-up primate brain, without new neurological systems
    - The difference from the brain of a mouse involves only a few modifications
    - The brain of a fish has almost all basic structures as our brain
  - Our larger-sized brain enabled the development of language, abstract reasoning, long-term planning, and other advanced cognitive abilities, superior to that of primates and other animals

The First Humans

MYA = Million Years Ago
KYA = Thousand Years Ago

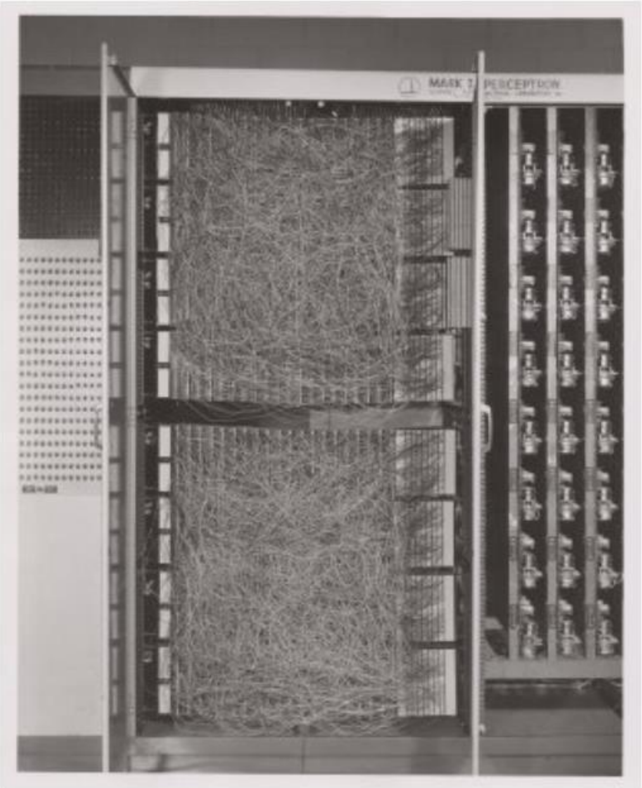


Source: Max Bennett (2023) A Brief History of Intelligence

AI Timeline: Early AI

AI Timeline

- 1943 – The first model of a simple artificial neuron proposed
- 1950 – Alan Turing introduced the Turing test
- 1955 – The term Artificial Intelligence used for the first time
- ***1956 – Workshop on AI held in Dartmouth College***, New Hampshire, organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, Claude Shannon
  - Official beginning of AI as academic discipline
  - A statement from the workshop proposal: "Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."
- 1958 – Perceptron algorithm proposed by Rosenblatt
  - Shown is the Mark I Perceptron computer, used for implementing the algorithm



AI Timeline: Early AI

AI Timeline

- 1959 – The term Machine Learning was used for the first time
- 1961 – The first industrial robot Unimate was installed on an assembly line at General Motors in New Jersey
- 1966 – Eliza, a chatbot that simulates conversations with a psychotherapist
- 1970-1980 – First AI winter, agencies reduced funding for AI projects due to unsatisfactory progress
- 1982 – An expert system was deployed for configuring computer orders
- 1987-1992 – Second AI winter, DARPA cut AI funding for expert systems
- 1990-1995 – The advent of machine learning and statistical methods
- 1997 – IBM's supercomputer Deep Blue won against world chess champion Garry Kasparov

AI Timeline: Modern AI

AI Timeline

- 2002 – Vacuum robots Roomba released
- 2006 – Facebook, Netflix, Twitter started using AI-based recommender systems
- 2011 – IBM's supercomputer Watson won against two human rivals in the quiz show Jeopardy

- 2012 – Deep NN model AlexNet won image classification contest: ***beginning of the era of deep learning***
- 2014 – GAN (Generative Adversarial Network) introduced
- 2016 – Google's DeepMind program AlphaGo defeated the Go grandmaster Lee Sedol
  - The game of Go is more difficult than chess, because the number of possible moves is much greater

AI Timeline: Recent AI

AI Timeline

- 2017 – Transformer network architecture was introduced in the paper by Vaswani et al. “Attention Is All You Need”
- 2018 – BERT language model establishing the pretrain-finetune paradigm for NLP
- 2020 – OpenAI’s GPT-3 is the first large language model with 175B parameters, performed well on many NLP tasks
- 2021 – DeepMind’s AlphaFold achieved high accuracy in predicting the 3-dimensional shape of proteins from amino acid sequences
- 2022 – OpenAI’s DALL·E 2 generated photorealistic images with remarkable quality
- 2022 – Facebook’s NLLB (No Language Left Behind) model for machine translation between 200 languages
- 2022 – OpenAI released ChatGPT, a large language model with human-like abilities in answering questions and chatting: ***beginning of Large Language Models revolution***

AI Timeline: Recent AI

AI Timeline

- 2023 – Google launched Gemini, Anthropic launched Claude 3: both reached the top spot on standards benchmarks at the time of launching
- 2024 – Vision-Language models (GPT-4 omni, LlaVA, Claude) for image interpretation and analysis introduced
- 2025 – Reasoning models trained to produce long chains of thought before answering become mainstream
- 2025 – Agentic frameworks enable LLMs to coordinate specialized agents for planning, research, and automation
- 2025 – Turing test passed by GPT-4.5, it was mistaken for human 73% of the time
- 2025 – Open AI o3 and Gemini Deep Think achieved gold-level medal performance on the International Mathematical Olympiad (IMO)
- 2026 – AI achieved the first perfect score (42/42) on IMO
- 2026 – AI produced an original mathematical discovery by disproving a mathematical conjecture in discrete geometry proposed by Erdos 80 years ago

DL Success in Computer Vision

Current AI Capabilities

- *Computer Vision* tasks
  - Image and video recognition/classification, segmentation, object detection, image synthesis
- Important architectures in CV
  - **AlexNet** (2012): Convolutional NNs for image recognition, 8 layers, GPU for parallel processing
    - Reduced the top-5 error on ImageNet from 26% by traditional ML approaches to 16%
  - **VGG, Inception** (2014): 16 to 22 layers CNNs, stacked 1x1 convolutions (Inception)
  - **ResNet** (2015): introduced residual connections, architectures with up to 152 layers
  - **Vision Transformers** (2020): employ attention layers, inspired by NLP transformers
  - **CLIP** (2021): trained on image-text pairs, enables zero-shot classification
  - **Segment Anything (SAM), DINOv2** (2023): vision foundation models for promptable image segmentation and unlabeled visual feature learning
  - **Vision-Language Models** (2024): vision encoders coupled to LLMs in a single model prompted in natural language

DL Success in Natural Language Processing

Current AI Capabilities

- ***Natural Language Processing*** (NLP) tasks
  - Text classification, text summarization, speech recognition, machine translation, dialog generation, part-of-speech tagging
- In the last decade, ***Large Language Models (LLMs)*** powered by transformer NNs achieved unprecedented success in NLP tasks
  - Training data for LLMs is a diverse set of text data gathered from the web
    - E.g., text collected from Wikipedia, e-books, news articles, and many other sources
  - Data preprocessing: text is split into individual word or sub-word units (tokens), and each token is mapped to a numerical vector in an embeddings space
  - Given a sequence of tokens from a dictionary, LLMs are trained to predict the next token (i.e., assign a probability to each token in the vocabulary)

Input: A quick brown	Output: fox
Input: Mary had a little	Output: lamb
Input: Nothing is	Output: impossible

Large Language Models

Current AI Capabilities

- ***Training LLMs*** involves several phases
  1. Pretraining
     - The model extracts knowledge from large text datasets consisting of trillions of tokens, based on next-token prediction
  2. Mid-training
     - The model is further trained on smaller, high-quality datasets containing content related to scientific knowledge, reasoning traces, coding, and long-context understanding
  3. Post-training
     a) Supervised fine-tuning (SFT)/instruction following - training on human-labeled prompt-response pairs
        - The model learns to generate responses that resemble the format of provided human examples

     b) Alignment/preference optimization - aligns the responses with human preferences for safety and helpfulness
        - Commonly used alignment methods include RLHF (Reinforcement Learning from Human Feedback) to rank responses, or DPO (Direct Preference Optimization) to optimize directly on preference pairs

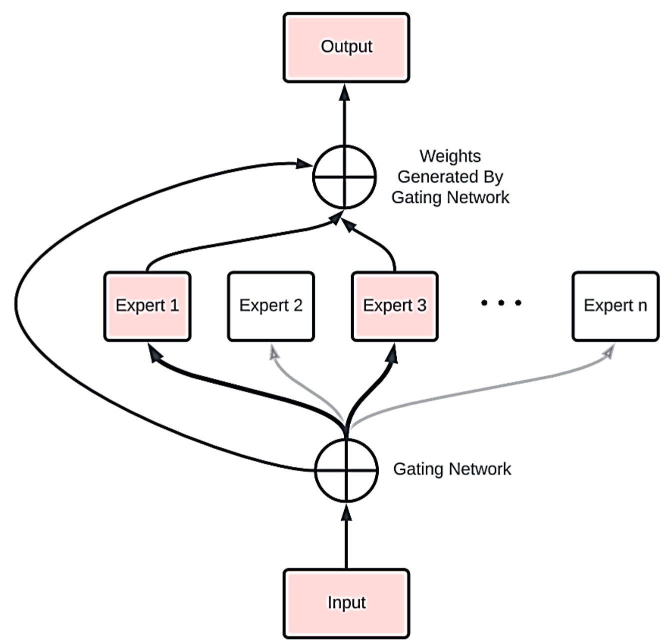
     c) Reasoning and agentic finetuning - improves reasoning, agentic and tool-use performance
        - For reasoning, Reinforcement Learning with Verifiable Rewards (RLVR) is applied to tasks such as math and logic problems or code, where the correctness of the responses can be automatically verified

     d) Safety and red teaming - improves robustness to adversarial prompts, bias mitigation

Large Language Models

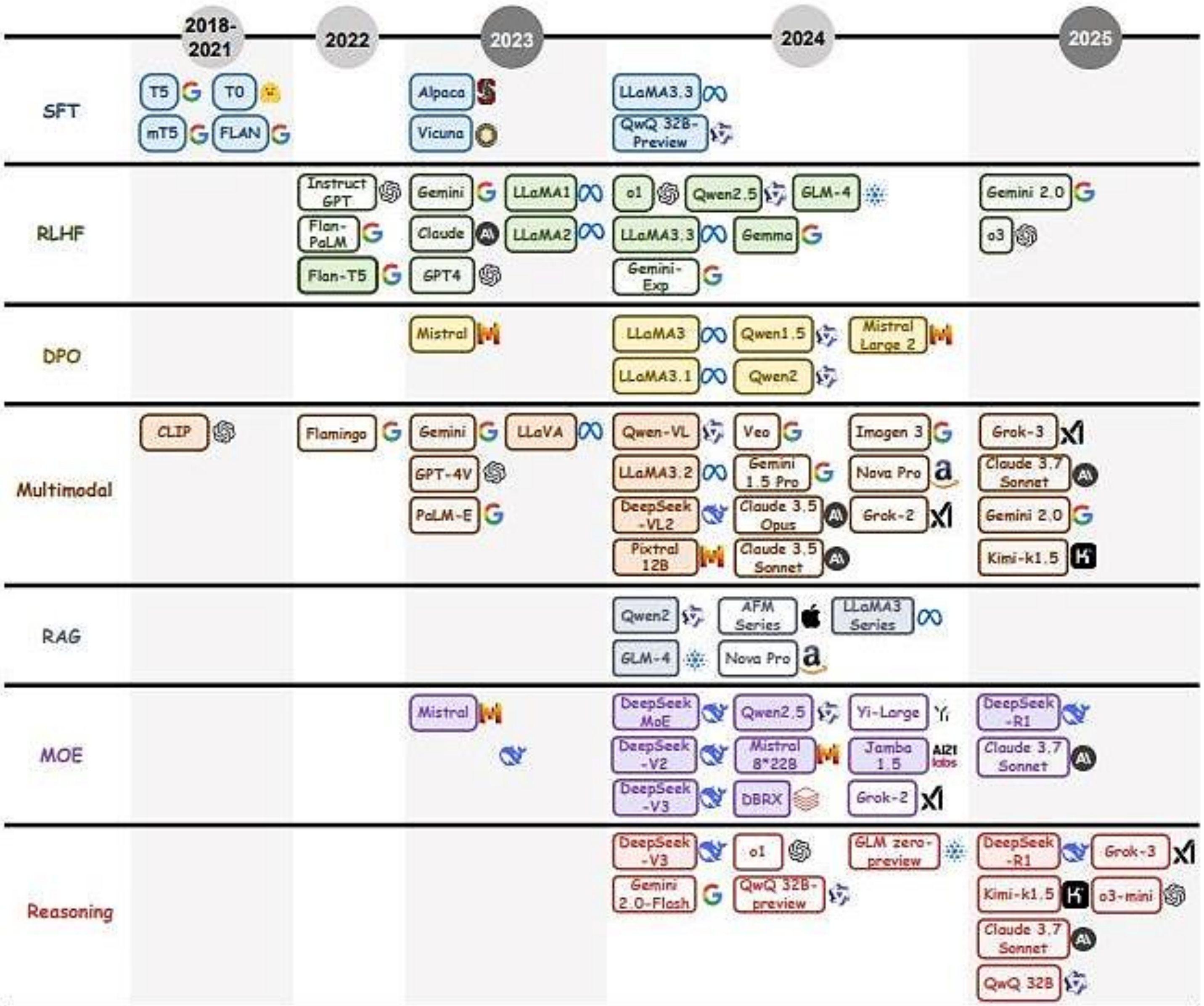
Current AI Capabilities

- Recent LLMs are very large deep neural networks having trillions of parameters
  - Frontier LLMs like GPT (OpenAI), Claude (Anthropic), and Gemini Pro (Google) reportedly contain between 2 and 10 trillions parameters
    - Note for comparison that human brain has between 100 and 500 trillion synaptic connections among approximately 86 billion neurons
- Training LLMs requires substantial ***computational resources*** and time: typically tens of thousands of GPUs are used for LLM training
    - E.g., purchasing cost of H200 GPU is about $40K -> cost of 10,000 H200 GPUs is about $400M
- The architecture of most frontier models is a ***Mixture-of-Experts (MoE)*** design
  - The network layers are split into many "expert" sub-networks that learn different aspects of a task
  - A gating network acts as a router and decides which expert(s) should process each token
  - MoE provides higher quality responses by training specialized experts, and allows efficient inference by activating only a few experts per token



Timeline of Large Language Models

Current AI Capabilities



LLM Scaling Laws

Modern AI Systems

- LLM *scaling laws* provide a relationship between the model's performance and model's scale (typically shown on a log scale)
  - Well known train-time compute scaling laws include Chinchilla and Kaplan laws (see left figure below)
    - They indicate that performance increases with more compute, tokens, and parameters
- OpenAI's test-time compute scaling law (right figure below) suggests that test-time compute follows the same trend as train-time compute
  - Performance improves by allowing LLMs to "think longer" during inference

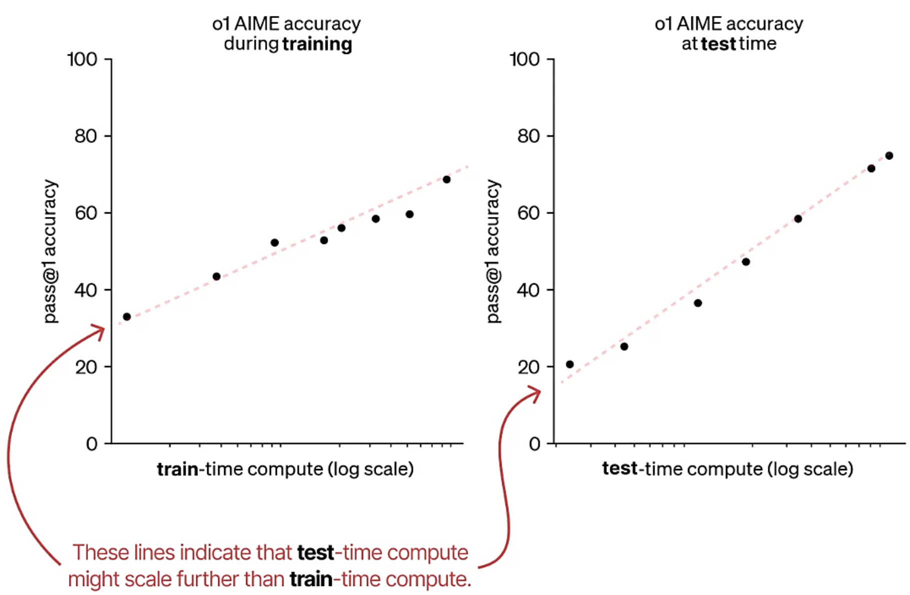


Figure from: A Visual Guide to Reasoning LLMs (link)

Reasoning LLMs

Modern AI Systems

- ***Reasoning LLMs*** perform multiple reasoning steps before providing an answer at test time
  - The reasoning steps or thought process break down the inference process into smaller, structured inferences (Chain-of-Thought)
    - The model can review its own work before committing to a final answer
  - Reasoning capability is typically trained with an RLVR phase, and it is today a default capability in most frontier LLMs

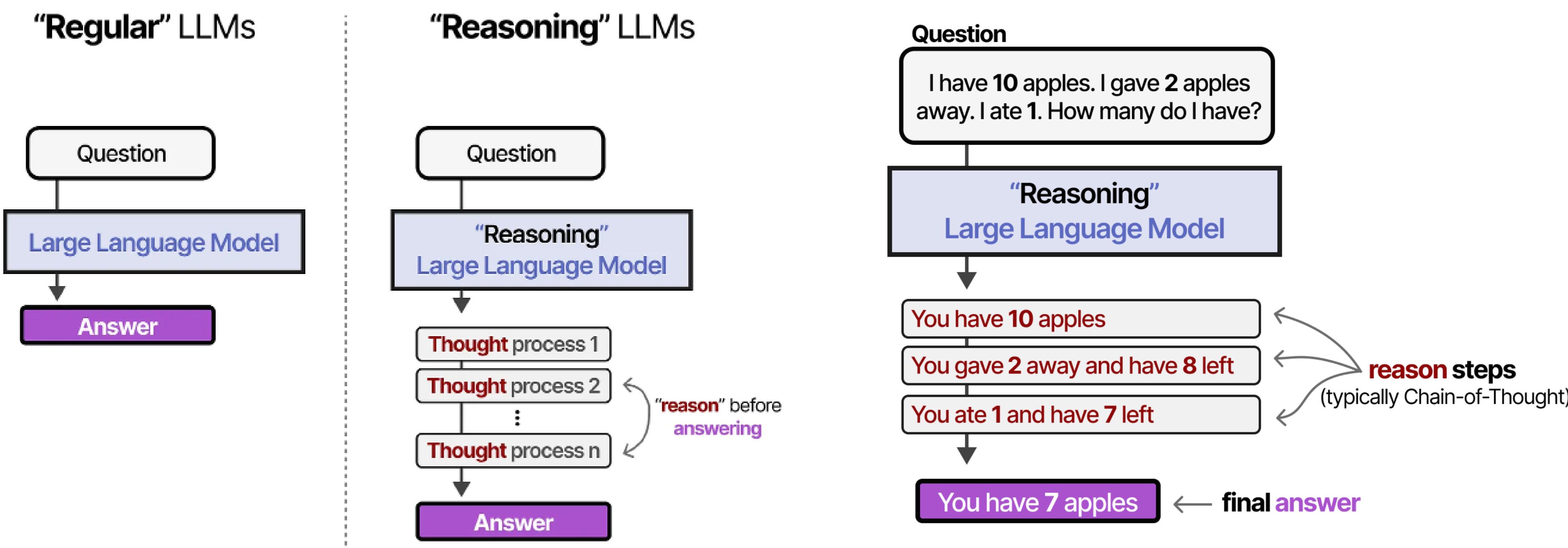


Figure from: A Visual Guide to Reasoning LLMs (link)

Evaluating and Comparing LLMs

Current AI Capabilities

- Evaluating and comparing LLMs is more challenging than classification DNNs
  - **Preference arenas** (LMArena): humans vote on LLM responses; this evaluation is sensitive to length, formatting, and tone of the responses
  - **Capability benchmarks** (MMLU, GPQA, SWE-bench, ARC-AGI): evaluate performance on fixed tasks (e.g., multiple-choice questions); this evaluation is sensitive to contamination (test data in the training set) and deliberate optimization for the benchmark (a.k.a. benchmaxing)
  - **Agentic benchmarks** (Terminal-Bench, OSWorld): evaluate multi-step tasks with tools; this evaluation is expensive and produces high-variance scores
- Benchmarks saturate quickly
  - MMLU, GSM8K, and HumanEval are effectively solved
  - AI performance on IMO problems went silver medal level in 2024, gold medal level in 2025, to perfect scores in 2026
- Questions to consider
  - Who ran the evaluation? The lab that built the model, or an independent party?
  - How much inference compute was allowed (since effort settings can change the score)?

LLMs Leaderboard by LMArena

Current AI Capabilities

Rank	Model	Net Improvement	Confirmed Success	Praise vs Complaint
1 1 ⟷ 7	Claude Fable 5 (High) Anthropic · Proprietary	▲ 12.58% ±2.19%	▲ 10.26% ±4.23%	▲ 24.42% ±7.82%
2 1 ⟷ 9	Claude Opus 5 (Max) Anthropic · Proprietary	▲ 11.88% ±2.81%	▲ 17.56% ±4.69%	▲ 25.63% ±9.98%
3 1 ⟷ 7	Claude Opus 5 (High) Anthropic · Proprietary	▲ 11.73% ±1.66%	▲ 16.60% ±3.11%	▲ 24.40% ±6.28%
4 1 ⟷ 11	GPT 5.6 Sol (xHigh) OpenAI · Proprietary	▲ 10.02% ±1.63%	▲ 8.61% ±3.32%	▲ 21.90% ±6.10%
5 1 ⟷ 10	Kimi K3 (Max) Moonshot · Kimi K3 license	▲ 9.91% ±1.19%	▲ 14.23% ±2.46%	▲ 17.45% ±4.26%
6 1 ⟷ 12	Claude Opus 4.8 (Thinking) Anthropic · Proprietary	▲ 9.43% ±1.64%	▲ 8.75% ±2.75%	▲ 21.05% ±5.25%
7 1 ⟷ 14	Claude Sonnet 5 (High) Anthropic · Proprietary	▲ 8.57% ±2.00%	▲ 7.95% ±3.98%	▲ 16.12% ±7.51%
8 3 ⟷ 13	GPT 5.5 (xHigh) OpenAI · Proprietary	▲ 8.49% ±0.95%	▲ 5.82% ±1.94%	▲ 12.91% ±3.38%
9 3 ⟷ 13	Claude Opus 4.7 (Thinking) Anthropic · Proprietary	▲ 8.19% ±1.26%	▲ 6.19% ±2.65%	▲ 11.20% ±4.39%
10 4 ⟷ 13	GPT 5.5 (High) OpenAI · Proprietary	▲ 7.89% ±0.88%	▲ 5.63% ±1.76%	▲ 11.31% ±3.13%
11 5 ⟷ 15	Claude Opus 4.7 Anthropic · Proprietary	▲ 7.37% ±1.31%	▲ 4.71% ±2.72%	▲ 12.28% ±4.45%
12 6 ⟷ 15	GLM 5.2 (Max) Z.ai · MIT · SiliconFlow	▲ 7.08% ±0.96%	▲ 9.08% ±1.94%	▲ 13.79% ±3.42%

Source: https://arena.ai/leaderboard/agent

Text-to-Image/Text-to-Video Models

Current AI Capabilities

- Text-to-image and text-to-video models produce images/videos with remarkable photorealism, accurate fine details, compositionality, spatial relations of the objects, and even with creativity in the synthesis
  - These models typically employ diffusion probabilistic networks, which learn the steps of adding and removing noise to images
- ***Text-to-image models***
  - DALL·E, GPT image (OpenAI)
  - Imagen, Nano Banana (Google)
  - Stable Diffusion (Stability.ai)
  - Midjourney (Midjourney.com)
- ***Text-to-video models***
  - Sora (OpenAI)
  - Veo (Google)
  - Dream Machine (Luma Labs)
  - Runway (RunwayAI)

Images Generated by DALL·E 2

Current AI Capabilities

- These are a few (cherry-picked) examples of images generated by DALL·E 2

A photo of a quaint flower shop storefront with a pastel green and clean white facade and open door and big window

Cat sipping tea and posting to twitter while sitting on a couch

A rabbit detective sitting on a park bench and reading a newspaper in a victorian setting

A lion in a hoodie hacking on a laptop

Teddy bears shopping for groceries in ancient Egypt

Teddy bears working on new AI research on the moon in the 1980s

Foundation Models

- ***Foundation models*** are large NN models trained at scale with high capabilities for transfer learning to many other applications
- The scale of these models results in new emergent capabilities, e.g., they perform well on tasks on which they were not explicitly trained to do
  - This allows adapting the models to new tasks with a small number of training data instances
    - E.g., few-shot learning refers to learning from only a few instances
- Notable applications of pretrained LLMs include:
  - Programming code completion models: CoPilot, Codex
  - Text-to-image generative models: DALL·E, Imagen, Stable Diffusion
  - Protein sequence prediction, solving math problems, preparing legal documents
- Transfer learning is what makes foundation models possible, but scale is what makes them powerful

AI Agents

- ***AI Agents*** use LLMs to understand instructions, reason through tasks, and take actions across digital tools or environments
- Abilities
  - Interpret natural language commands, plan multi-step solutions, call external tools (e.g., web search, code execution), adapt behavior based on feedback
- Unlike a standard LLM that uses a single pass to produce an output, an agent operates within a continuous loop
  - The loop involves: observing the environment, reasoning about the current state, formulating a plan, executing actions, and using the output of those actions to refine the next steps
- Task example: "Plan a marketing campaign for next quarter"
  - The agent breaks the task into smaller steps, uses external tools like web search or databases, checks the results, and iterates until complete

Evolution of AI Capabilities

Modern AI Systems

- **Pattern recognition AI period**
  - Analyze existing data, make predictions
    - Classification models for identifying spam emails, recognizing handwritten digits
    - Regression models for predicting house prices, stock trends
- **Generative AI period**
  - Generate new content
    - LLMs for generating human-like text and conversations
    - Text-to-image models for creating realistic images from text prompts
    - Models for composing music, writing code, creating videos
- **Task execution AI period**
  - AI agents perform actions
    - Booking flights and making reservations
    - Analyzing documents and writing reports
    - Controlling software applications
- Next AI period (?) – the last slide provides suggestions

Concerns Regarding LLMs

AI Limitations and Challenges

- Misuse and unethical use
  - Amplifying disinformation, generating synthetic media at scale, enabling fraud and impersonation, privacy risk with sensitive information in training data
- Environmental impact and power consumption
  - LLM training and inference cause high carbon emissions
  - Power demand for data centers impacts the electrical grid; water use for cooling is becoming an important constraint
- Unreliable outputs
  - Hallucinations (generating false information), bias inherited from training data, inconsistent outputs on repeated queries, sycophancy (LLMs agree with the user and abandon answers on pushback)
- Computational resources and centralization of power
  - Training frontier LLMs costs hundreds of millions of dollars and is concentrated in a few labs, which increases inequality and limits access
- Governance
  - Current AI governance is based on a patchwork of state laws and sectoral rules, without comprehensive federal legislation

Engineering vs Science Phase of Technology

AI Limitations and Challenges

- Theoretical guarantees about AI performance are lacking at the present time
  - Currently, AI is in ***Engineering phase***: models are designed to solve tasks, are integrated into new products, add value to companies, have economic impact
  - AI is expected to progressively reach ***Science phase*** as new theoretical frameworks emerge that explain model behavior, establish convergence and stability guarantees, and reliably predict successes and failures of learning algorithms
- Various technologies historically began with an engineering phase (inventions made, products built) to be later followed by a science phase (theory developed)
  - Steam engines were used in paper mills and factories since 1776; the theory of Thermodynamics was developed between 1820s and 1850s
  - Airplanes were constructed and flown since 1903; the modern theory of Aerodynamics was developed in 1930s
  - Electric circuits were discovered around 1800; the theory of Electromagnetism was founded between 1820s and 1830s

Trustworthy AI

- ***Trustworthy AI*** – efforts to address the limitations to ensure that end-users can trust the predictions by AI models
- Topics in trustworthy AI include:
  - Robustness
    - Even unnoticeably small perturbations can impact the model predictions
  - Generalization
    - OOD (out-of-distribution) inputs; e.g., a model trained on medical images in one hospital performs poorly on images in another hospital (due to different equipment or settings used)
  - Explainability
    - The decision-making process of large models is non-transparent and difficult to understand
  - Fairness
    - Predictions can show bias against demographic groups, based on gender, age, culture
  - Privacy protection
    - Models can memorize and reveal input data; e.g., a model can reveal sensitive private information in medical records used for training
  - Ethics
    - The models should produce ethical decisions that are aligned with our human values (also referred to as AI Alignment)

Moravec's Paradox

AI Limitations and Challenges

- The paradox is based on observations by the researcher Hans Moravec: the hard problems for humans are easy for computers, and the easy problems for humans are hard for computers
  - E.g., it is easier for machines and computers to play chess or solve logic problems, than to make a coffee or plant a tree
  - Robotic tasks that involve sensorimotor and perception skills are more difficult than planning and reasoning tasks
- One possible explanation of Moravec's paradox is because sensorimotor skills are older and developed earlier in the evolution than reasoning skills
  - We can perform skills like recognizing a face, moving in space, or catching a ball unconsciously and effortlessly, because they have been optimized over millions of years
  - Abstract thinking, reasoning, or solving math problems are thousands of years old skills that we haven't mastered completely, and we need to put more effort
- We should expect that the difficulty to automate a skill with AI is proportional to the amount of time it took for the skill to develop in animals and humans

The Bitter Lesson

AI Limitations and Challenges

- ***The Bitter Lesson*** (2019) is a short paper by Rich Sutton
  - http://www.incompleteideas.net/IncIdeas/BitterLesson.html
- The Bitter Lesson is based on his observations regarding the historical development of AI methods, which can be characterized with three phases:
  - Phase 1 - AI researchers incorporate human domain knowledge into their AI methods, which helps in short term
  - Phase 2 - In the long term, the performance of such models plateaus without further progress
  - Phase 3 - Progress is eventually achieved by general methods that scale computation with search and learning
- In conclusion:
  - AI methods that leverage computation and search at scale are the most effective
  - Human-centric approaches complicate methods and make them less suited to leveraging computation and search at scale
  - The search for new solutions should be done by our methods, not by us

Prospective Trends in AI

Prospective Trends in AI

- *Multi-agent AI systems*
  - Specialized AI agents that collaborate with other agents, use external tools, and autonomously execute complex workflows
- *World models and simulation*
  - AI frameworks that build internal representations of physical and digital environments to improve planning, prediction, and decision-making
- *Continual and lifelong learning*
  - Models that continuously learn from new experiences and adapt over time to changing environments
- *Embodied AI and robotics*
  - Integration of foundation models with robots, autonomous vehicles, and industrial systems to advance perception, manipulation, and real-world interaction
- *Scientific discovery and autonomous research*
  - AI systems that assist or autonomously perform scientific research and discoveries in mathematics, medicine, and engineering